Evolving landscape of treatments targeting the microenvironment of liver metastases in non-small cell lung cancer

Lingling Zhu[1,2], Xianzhe Yu[3], Xiaojun Tang[1], Chenggong Hu[4], Lei Wu[5], Yanyang Liu[6], Qinghua Zhou[1]

[1]Lung Cancer Center, Lung Cancer Institute, West China Hospital, Sichuan University, Chengdu, Sichuan 610041, China;
[2]Key Laboratory of Drug Targeting and Drug Delivery Systems, Ministry of Education, West China School of Pharmacy, College of Polymer Science and Engineering, Sichuan University, Chengdu, Sichuan 610041, China;
[3]Department of Gastrointestinal Surgery, Chengdu Second People's Hospital, Chengdu, Sichuan 610041, China;
[4]Department of Critical Care Medicine, West China Hospital, Sichuan University, Chengdu, Sichuan 610041, China;
[5]Core Facility of West China Hospital, Sichuan University, Chengdu, Sichuan 610041, China;
[6]Department of Medical Oncology, Cancer Center, West China Hospital, Sichuan University, Chengdu, Sichuan 610041, China.

Abstract
Liver metastases (LMs) are common in lung cancer. Despite substantial advances in diagnosis and treatment, the survival rate of patients with LM remains low as the immune-suppressive microenvironment of the liver allows tumor cells to evade the immune system. The impact of LMs on the outcomes of immune checkpoint inhibitors in patients with solid tumors has been the main focus of recent translational and clinical research. Growing evidence indicates that the hepatic microenvironment delivers paracrine and autocrine signals from non-parenchymal and parenchymal cells. Overall, these microenvironments create pre- and post-metastatic conditions for the progression of LMs. Herein, we reviewed the epidemiology, physiology, pathology and immunology, of LMs associated with non-small cell lung cancer and the role and potential targets of the liver microenvironment in LM in each phase of metastasis. Additionally, we reviewed the current treatment strategies and challenges that should be overcome in preclinical and clinical investigations. These approaches target liver elements as the basis for future clinical trials, including combinatorial interventions reported to resolve hepatic immune suppression, such as immunotherapy plus chemotherapy, immunotherapy plus radiotherapy, immunotherapy plus anti-angiogenesis therapy, and surgical resection.
Keywords: Non-small cell lung cancer; Liver metastasis; Combination therapy; Immune checkpoint inhibitors; Immune tolerance; Immunotherapy

Introduction

The incidence of lung cancer remains high, making it one of the most frequently diagnosed malignancies. Non-small cell lung cancer (NSCLC) accounts for approximately 85% of lung cancer cases. Lung adenocarcinoma (LUAD) and squamous carcinoma of the lungs are the most prevalent subtypes of NSCLC.[1] Fifty-seven percent of patients with lung cancer are diagnosed with metastatic disease, and their 5-year survival rate is 6%.[2]

Liver is a common site of NSCLC metastasis, affecting approximately 20% of NSCLC patients.[3] Liver metastasis (LM) can modulate systemic immune responses,[4] leading to a poor prognosis. Patients with LM demonstrate a median overall survival (OS) of 3–4 months,[5] which is worse than that of patients with metastases in other organs, including the brain (7 months), bone (6 months), and lymph nodes.[6,7] For LMs associated with NSCLC, treatment includes surgical resection, transarterial chemoembolization, radiofrequency ablation, molecularly targeted agents (targeting the mutated epidermal growth factor receptor [EGFR], rearranged anaplastic lymphoma kinase [ALK], and rearranged reactive oxygen species [ROS] proto-oncogene 1, receptor tyrosine kinase), palliative radiotherapy, and immunotherapy.[8] Patients with LMs who undergo targeted or cytotoxic therapy[9] or surgery[10] have poor prognoses. Moreover, LMs negatively affect the responses to immunotherapy.[11] For instance, pembrolizumab,[12] nivolumab,[13] or durvalumab[7] is

Access this article online

Quick Response Code:

Website:
www.cmj.org

DOI:
10.1097/CM9.0000000000002981

Lingling Zhu and Xianzhe Yu contributed equally to the manuscript.

Correspondence to: Qinghua Zhou, Lung Cancer Center, Lung Cancer Institute, West China Hospital, Sichuan University, No. 37 Guo Xue Xiang, Wuhou District, Chengdu, Sichuan 610041, China
E-Mail: prof_qh_zhou@126.com;
Yanyang Liu, Department of Medical Oncology, Cancer Center, West China Hospital, Sichuan University, No. 37, Guo Xue Xiang, Wuhou District, Chengdu, Sichuan 610041, China
E-Mail: lyy3346@126.com

Chinese Medical Journal 2024;137(9)

Received: 05-12-2023; **Online:** 22-01-2024 **Edited by:** Xiangxiang Pan and Peifang Wei

significantly less effective in NSCLC patients with LMs than in those without LMs.

Generally, immune responses to intrahepatic tumor antigens result in the systemic inhibition of antitumor immunity.[12] Cancer cells that enter the liver are exposed to a complex and distinct microenvironment (dendritic cells [DCs], Kupffer cells [KCs], and sinusoidal vessels with immunosuppressive functions) that may contribute to tumor evasion of immune surveillance during the course of immunotherapy. Hepatic parenchymal and non-parenchymal cells, together with recruited immune and inflammatory cells, are involved in the response to invasive tumor cells and may favor or suppress the progression of metastasis.[14] Compared with conventional antigen-presenting cells (APCs) in the steady-state liver environment, such as DCs, unconventional APC populations, constituting KCs, liver sinusoidal endothelial cells (LSECs), hepatic stellate cells (HepSCs), and hepatocytes, express low levels of major histocompatibility complex (MHC)-I/MHC-II and co-stimulatory molecules.[15] The liver has unique immunological properties, leading to a systemic loss of T cells, including T-cell anergy or apoptosis, regulatory T-cell (Treg) induction, or effector T-cell elimination mediated by conventional and unconventional APCs.[16]

The pivotal contribution of the liver microenvironment to the promotion of cancer cells and their growth within the liver suggests that targeting molecules and cells in the liver microenvironment is a preventive and therapeutic strategy for LM.[17] Targeting the factors of the tumor microenvironment (TME) alone or in combination with tumor-directed biotherapy or chemotherapy is advantageous based on the following reasons: (1) cancer cells rely on a supportive microenvironment for survival and growth, (2) the TME is composed of genetically stable cells that have more predictable responses and characteristics, and (3) targeting the TME may be advantageous for various cancer types, particularly for tumors that predominantly metastasize to the same location, for instance, the liver.[17]

This review provides insights into the epidemiology, physiology, pathology and immunology of LMs in patients with NSCLC. We focused on investigating the mechanisms by which certain etiologies alter hepatic homeostasis and addressed how such alterations facilitate tumor metastasis and modulate the surveillance of primary tumors or aggressive metastatic cells. We also explored the rationale and strategies for targeting the liver microenvironment to prevent and treat LMs.

Epidemiology and Prognosis of NSCLC-related LMs

The three most common organs involved in NSCLC metastasis are the adrenal glands, liver, and lungs.[4] LMs occur in approximately 4–20% of patients with NSCLC and 17% of patients with SCLC.[3,9,18] Data derived from the Surveillance, Epidemiology, and End Results database indicate that 5.1% of patients present with synchronous LMs upon the diagnosis of primary cancer.[19]

LM is a major factor related to poor progression-free survival (PFS) and OS among patients with NSCLC.[4] For instance, a previous cohort study of 1096 patients with NSCLC and LMs conducted between 2006 and 2014 reported a 6-month survival rate of 68.2%.[20] The reported median OS of patients with and without LMs who received nivolumab treatment ranged from 3.6 to 7.5 months and 5.8 to 20.7 months, respectively.[21] Among patients with chemotherapy-naïve, advanced *EGFR*-mutated NSCLC without brain metastasis, the median PFS was markedly poorer in those with LMs than in those without LMs (5.1 months and 12.9 months, respectively).[22] A single-center study involving 41 patients reported immune checkpoint inhibitor (ICI) therapy response rates of 22.5% and 29.7% in patients with and without LMs, respectively ($P = 0.43$).[23]

Pathophysiological Mechanisms in LM of NSCLC

Anatomical features

The liver receives its blood supply from two sources, namely two-thirds to three-fourths of the blood flows from the portal vein, while the remainder flows from the hepatic aortic system. The hepatic artery is the primary route by which tumor cells enter the liver,[24] while lung cancers seed tumor cells into the liver via the portal vein.[3] The specific anatomical localization, immunosuppressive environment, and endothelial cell infiltration of the liver make it susceptible to tumor metastasis from extrahepatic cancers, referred to as secondary liver cancers.[25] For example, the cluster of differentiation 8 positive ($CD8^+$) T cells targeting antigens expressed in the liver may exhibit impaired function (T-cell failure) when continuously exposed to a high antigen load within the liver after their initial activation in the secondary lymphoid organs,[25] thus providing the immunosuppressive microenvironment for LMs.

Genomic aberrations

LM can arise from genomic events in the cancer cells, such as the activation of oncogenes and inactivation of tumor suppressor genes. For instance, USP22, a ubiquitin hydrolase, enhances the proliferation, angiogenesis, and recurrence of NSCLC.[26] *USP22*-knockout is a novel NSCLC treatment strategy that may activate a wide range of anticancer activities by affecting multiple signaling pathways that inhibit angiogenesis, proliferation, epithelial-to-mesenchymal transition (EMT), and the expression of *KRAS* and c-Myc.[26] In lung cancer metastasis models *in vivo*, *USP22*-knockout cancer cells generated substantially smaller and fewer LMs.[26]

The expression of 5′-nucleotidase domain-containing 2 (NT5DC2), an NT5DC family member, is increased in human NSCLC tissues.[27] Mechanistically, the proliferation and apoptosis of NSCLC are regulated by NT5DC2 in a p53-dependent manner. NT5DC2 suppression triggers both cell cycle arrest and apoptosis in NSCLC cells in the G2/M phase, while the deletion of *NT5DC2* inhibits EMT and NSCLC-related LM.[27]

The inhibitor of differentiation (ID) family of proteins encourages tumor growth, angiogenesis, and metastasis.[28] ID1 is modulated via the c-Jun N-terminal kinase (JNK) pathway by *KRAS* oncogenes, and deletion of *ID1* leads to the downregulation of mitogenic machinery elements through the suppression of the transcription factor FOS-like 1, together with several kinases in the *KRAS* signaling network. The expression of ID1 in the liver microenvironment and lung cancer cells promotes LM by increasing the colonization and migration ability of lung cancer cells, regulating the EMT phenotype, and establishing the premetastatic niche (PMN).[28] The abolition of ID1 induces G2/M phase arrest and apoptosis in *KRAS*-mutated LUAD.[27] *Id1* deletion in xenograft models markedly compromises the growth and maintenance of tumors, together with LMs, thereby improving survival.[29]

Adenylate kinase 4 (AK4) is a biomarker in lung cancer metastasis,[30] the overexpression of which augments the hypoxia-inducible factor-1α (HIF-1α) protein expression by augmenting intracellular ROS levels and subsequently triggers EMT and invasion under hypoxic conditions. In a mouse model, AK4 overexpression reduces hypoxic necrosis and promotes LM.[30] Reducing ROS production with N-acetylcysteine eliminates the AK4-induced invasion potential under hypoxic conditions. Thus, the AK4–HIF-1α signaling axis might be a promising treatment target for the treatment of LM from lung cancer.

Collectively, the above-mentioned molecules, including USP22, ID1, NT5DC2, and AK4, can modulate the expression of proliferation-, angiogenesis-, and EMT-related proteins in either cancer cells or in the hepatic microenvironment. This modulation leads to the effective liver colonization of cancer cells and secondary metastatic outgrowth.[26–28,30,31] This finding underscores the potential of targeting EMT as a promising strategy for inhibiting LM.

Immunosuppressive microenvironments

Several reliable preclinical models have been used to evaluate the hepatic metastasis of tumors, including implantation mouse models,[27,32–34] *ex vivo* liver model systems,[35–37] genetically engineered mice,[38] and patient-derived xenografts.[38] Although these models present both merits and shortcomings[39–41] [Supplementary Table 1, http://links.lww.com/CM9/B859], two key ecological niches promoting the LM process have been elucidated using a cellular and hepatic metastasis mouse model of NSCLC. First, the TME at secondary organ sites can favorably influence metastatic growth before tumor cell entry, which is termed the PMN. A PMN is created by molecules secreted from the primary tumor that recruit non-tumor cells such as KCs, HepSCs, myeloid-derived suppressor cells (MDSCs), endothelial cells, and neutrophils, as well as extracellular matrix molecules, including collagen I, II, and IV; FAT atypical cadherin 1 (FAT1); Wnt/β-catenin; S100 in the liver.[18] The PMN is formed early, even before tumor cells enter the liver.[42] Second, the post-invasive tumor ecological niches that develop after the entry of tumor cells can be described in four critical phases: the microvascular phase, which involves the invasion of cancer cells and their arrest in the sinusoidal vessels; the pre-angiogenic stage outside the vessels; the angiogenic phase that supplies cancer cells with nutrients and oxygen; and finally, the phase in which LMs expand and grow into clinically detectable tumors.[18] These various cell types, along with the growth factors and chemokines/cytokines they secrete and the molecular signaling involved, represent potential targets for preventing and treating LM [Figures 1 and 2].

Role of liver-specific cells

LSECs, KCs, HepSCs, parenchymal hepatocytes, DCs, and resident natural killer (NK) cells all correlate with the promotion and maintenance of LMs, which mainly induce the expression of immune cells, such as neutrophils, macrophages, and monocytes, among others.[18]

Neutrophils

Neutrophils, which are part of an innate immune response to pathogens, can also be activated promptly by invading cancer cells. The number of neutrophils present in peripheral blood correlates positively with LM in patients with lung cancer.[43] In a murine model, the interaction between circulating tumor cells and adherent neutrophils in the inflamed sinusoids may be conducive to the arrest of Lewis lung carcinoma (LLC) cells in the liver.[44] Neutrophils promote cancer cell adhesion, mediated by neutrophil membrane-activated complex 1 (Mac-1)/intercellular adhesion molecule-1 (ICAM-1) within the liver sinusoids, thereby facilitating LM.[45] However, atypical chemokine receptors are leukocyte trafficking, immunity, and inflammation modulators.[46] In an orthotopic mouse

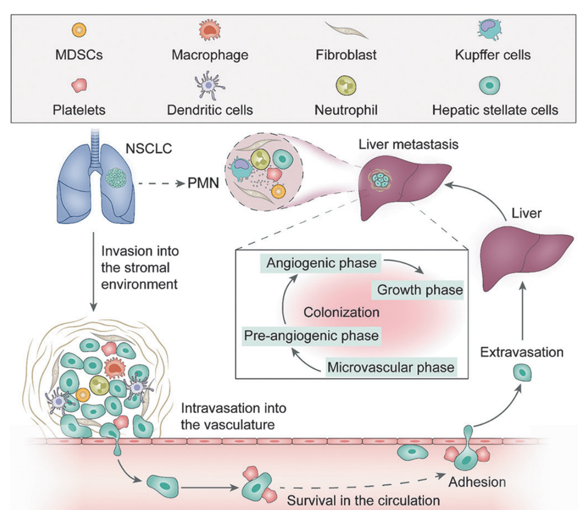

Figure 1: Phases of NSCLC liver metastases. Liver metastases develop after the hematogenous spread of NSCLC cells from the lungs to the liver, which provides "fertile soil" to develop metastasis. First, tumor cells invade the surrounding tissues, venules, capillaries, lymphatic system (intravasation), and circulation, and survive in the circulation; subsequently, the tumor cells adhere to the vascular wall and extravasate from the hepatic vasculature. After extravasation, the tumor cells experience four pivotal phases, including the microvascular, pre-angiogenic, angiogenic, and growth phases, causing micrometastases; later colonize and form macroscopic liver metastases. MDSCs: Myeloid-derived suppressor cells; PMN: Premetastatic niche; NSCLC: Non-small cell lung cancer.

model with 4T1 mammary carcinoma, *ACKR2*-deficient neutrophils showed an accelerated maturation rate, thus preventing neutrophil-mediated metastasis.[46] Therefore, neutrophils possess both anti- and pro-metastatic functions based on disease context.[47]

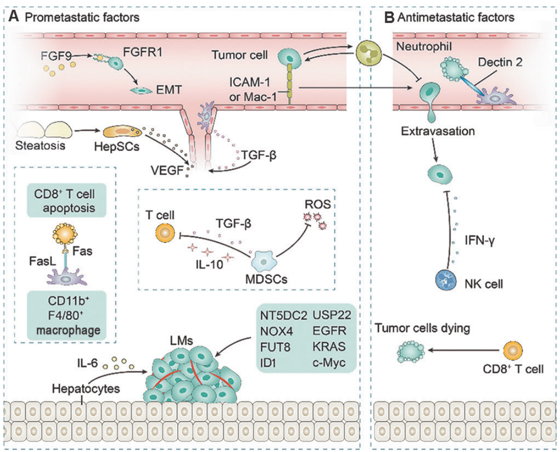

Figure 2: Pro-metastatic and anti-metastatic microenvironment of the liver in NSCLC. (A) FGF9 promotes EMT by interacting with FGFR1 on Lewis lung cancer cells.[49] Steatosis-activated HepSCs promote angiogenesis and metastasis during LM development.[59] Hepatic monocyte-derived CD11b$^+$F4/80$^+$ macrophages lead to antigen-specific T-cell apoptosis through the Fas–FasL pathway.[4] MDSCs that generate immunosuppressive cytokines, such as TGF-β and IL-10, express arginase to inhibit the proliferation of T cells and ROS generation.[50] Hepatocytes activate transcriptional activator 3 signaling and signal transducers depending on IL-6.[55] Neutrophils promote cancer cell adhesion mediated by neutrophil Mac-1/ICAM-1 within liver sinusoids and facilitate LM.[43] The molecules and signaling pathways involved in LM include NT5DC2, USP22, NOX4, EGFR, FUT8, KRAS, ID1, and c-Myc. (B) Activated CD8$^+$ T cells induce NSCLC cell apoptosis in the liver. Neutrophils prevent metastasis mediated by an accelerated neutrophil maturation rate and anti-metastatic activity.[44] NK cells present anti-metastatic tumor ability following IFN-γ induction.[54] Dectin-2 mediates the uptake and clearance of cancerous KCs.[48] CD: Cluster of differentiation; EGFR: Epidermal growth factor receptor; EMT: Epithelial-to-mesenchymal transition; FGF9: Fibroblast growth factor 9; FGFR1: Fibroblast growth factor receptor 1; FUT8: Fucosyltransferase 8; HepSCs: Hepatic stellate cells; Id: Inhibitor of differentiation; IFN-γ: Interferon-γ; ICAM-1: Intercellular adhesion molecule-1; IL: Interleukin; KCs: Kupffer cells; LM: Liver metastasis; Mac-1: Membrane-activated complex 1; MDSCs: Myeloid-derived suppressor cells; NK: Natural killer; NOX4: NADPH oxidase 4; NSCLC: Non-small cell lung cancer; NT5DC2: 5′-Nucleotidase domain containing 2; ROS: Reactive oxygen species; TGF-β: Transforming growth factor-β.

Macrophages

Macrophages are associated with liver metastasis from lung cancer. Chang *et al*[48] found that intratumor injections of fibroblast growth factor 9 (FGF9) promote EMT, infiltration of M2-macrophage into the TME, and LM via its interaction with fibroblast growth factor receptor 1 (FGFR1) in a subcutaneous LLC-bearing mouse model. FGF9 stimulates LM and LLC tumorigenesis by activating the extracellular signal-regulated kinase (ERK), protein kinase (AKT), and focal adhesion kinase (FAK) signaling via the FGFR1.

KCs account for approximately 10% of hepatocytes and are the largest tissue-resident macrophage population,[49] and critically mediate the immunosuppressive liver microenvironment through the expression of cytotoxic T-lymphocyte antigen-4 (CTLA-4), programmed cell death ligand 1 (PD-L1), transforming growth factor-β (TGF-β), or interleukin (IL)-10, and reduction of the levels of co-stimulatory molecules (CD80 and CD86), thereby activating Tregs.[50] Targeting KCs might be a promising strategy for preventing the development of incipient LMs. KC depletion by gadolinium chloride results in a decreased liver tumor burden. Dectin-2, a unique C-type lectin selectively expressed by DCs, mediates cancer cell uptake and clearance by KCs.[51] Kimura *et al*[51] reported that dectin-2-mediated cancer cell phagocytosis by KCs in an *in vivo* LUAD mouse model demonstrated LM suppression. Mice with LMs exhibited increased expression of CD11b$^+$F4/80$^+$ myeloid cells along with a decreased expression of CD4$^+$ T cells. Mechanistically, the liver monocyte-derived CD11b$^+$F4/80$^+$ macrophages caused antigen-specific T-cell apoptosis via the FAS–Fas ligand (FASL) pathway.[4]

MDSCs

Another heterogenic immature myeloid cell with potent immunosuppressive function is the MDSC. MDSCs generate immunosuppressive cytokines, such as TGF-β and interleukin (IL)-10, and express arginase, which inhibits the proliferation of T cells and ROS generation.[52] Therefore, immunosuppressive myeloid cells, such as MDSCs and M2 macrophages functioning as main barriers against effector lymphocytes, are potential targets for immunomodulatory drugs.

T cells

The presence of LMs is related to fewer CD8$^+$ T cells at the margins of distant tumor infiltration, a cellular characteristic that correlates with a response to programmed cell death protein 1 and its ligand (PD-1)/PD-L1 inhibitors.[53] Eliminating antigen-specific CD8$^+$ T cells results in systemic immunosuppression.[4] Recent studies examining the relationship between LM and treatment response in patients with lung cancer revealed a decreased density of CD8$^+$ T cells at the margins of invasive tumors in LM biopsies, which may explain the poor survival.[53] Several mechanisms have been proposed to account for an immune tolerance-induced liver, such as the activation of Tregs by KCs.[54] Tregs suppress abnormal self-antigen and antitumor immune responses.[12]

DCs

Compared with DCs from other tissues, liver DCs poorly affect T-cell activation due to their immaturity and the high IL-10 and low IL-12 levels in the hepatic microenvironment.[55] The hepatic microenvironment enables the differentiation of hematopoietic progenitor cells into regulatory DCs that sustain hepatic tolerance.[50]

NK cells

As a heterogeneous population, hepatic resident NK cells exert an essential role in infection and tumor immunosurveillance. The liver participates in lung metastasis as a leukocyte supplier.[56] Liver-derived leukocytes display liver-like characteristics and are designated hepato-entrained leukocytes, which contain B220$^+$CD11c$^+$NK1.1$^+$

cells with antimetastatic tumor ability following interferon-γ induction.

Hepatocytes

Hepatocytes are the bases of the liver and comprise approximately 60% and 80% of the liver's cell number and mass, respectively.[3] Although data related to the effect of hepatocytes on LM progression remain scarce, available studies have reported that hepatocytes coordinate the accumulation of bone marrow cells and fibrosis, helping to create a microenvironment conducive to metastasis.[57] In mice displaying early pancreatic tumorigenesis, hepatocytes activate the signal transducers and activators of transcription 3. This process depends on the release of non-malignant cell-derived IL-6 into the circulation and enhances the production of serum amyloid A1 and A2.[57]

LSECs

As the initial barrier to cancer cell invasion, LSECs are a source of cytokines for the recruitment of immune cells following activation by invading cancer cells.[58] The specific characteristics of the endothelial cell surface may play an essential role in determining metastatic patterns.[59] This finding may be attributed to the differences in tumor cell adhesion propensity along with the preference for various endothelial cells. Disruption of the interaction between tumor cell-derived selectin ligands and endothelial cell selection reduces the recruitment of LLC cells to the liver.[44]

HepSCs

HepSCs account for approximately 15% of non-parenchymal hepatocytes and coordinate the fibrotic response of the liver (e.g., wound healing). Hepatic steatosis, which causes liver inflammation and fibrosis, is an individual predictor of LM in patients with NSCLC.[60] Mechanistically, steatosis-activated HepSCs impact the progression and promotion of metastasis and organize angiogenesis during LM development.[61]

Cell interaction mediated by cytokines and chemokines

Growth factor and chemokine receptors are the most intriguing emerging targets in the TME as they are responsive to pharmacotherapy using antibody- and small molecule-based strategies. Preclinical data have supported their usefulness as therapeutic targets for LMs. Infectious complications have been associated with an increased risk of metastasis following tumor resection.[62] The innate immune system can be activated with lipopolysaccharide (LPS). Toll-like receptor 4 recognizes LPS, a gram-negative bacterial cell wall component. In C57BL/6 mice, gram-negative *Escherichia coli*-related pneumonia enhances LM in murine H59 NSCLC via the activation of Toll-like receptor 4.[62]

TGF-β, produced by tumor cells or the surrounding stromal cells,[63] is a primary driver of the fibrotic and immunosuppressive microenvironment and is critical for angiogenesis and growth of LM.[64] TGF-β1 bound to the membranes of MDSCs is the primary negative regulator of hepatic NK cells, inhibits the proliferation of T cells, and induces Tregs in tumor-bearing hosts.[52] TGF-β signaling by monocyte myeloid cells inhibits $CD8^+$ T cell activity during lung metastasis and promotes tumor growth in LM.[65] Cells pretreated with TGF-β1 tend to metastasize to the liver in experimental lung cancer metastasis mouse models.[66] Meanwhile, blocking TGF-β triggers a potent and lasting cytotoxic T-cell activity against cancer cells and prevents metastasis.[67] Therefore, TGF-β is one of the central cytokines in LM,[68] and future studies should focus on investigating TGF-β inhibitory strategies.

The complex interplay of chemokine ligand–receptors among different cells further complicates the signal transduction cascade in LM, inducing tumor angiogenesis, EMT, and migration/invasion.[69] For instance, tumor-derived chemokines C-C motif chemokine ligand 5 (CCL5) and CCL7, as type IV collagen regulatory genes, facilitate LM of lung carcinoma ($M27^{colIV}$) cells via FAK signaling.[70] Autocrine CCL7/C-C motif chemokine receptor 3 (CCR3) signaling functions downstream of collagen IV/integrin α2/FAK signaling and promotes EMT and cancer cell migration. This signaling is conveyed via the MEK/ERK and phosphoinositide 3-kinase (PI3K) pathways and involves nuclear factor kappa B (NF-κB) activation downstream of AKT. Moreover, CCL5/CCR5 signaling regulates the T-cell response in the TME.[70]

Cell–cell communication mediated by exosomal substances in LMs

Exosomes, a subset of extracellular vesicles (EVs), contribute to building a supportive microenvironment for LMs[71] and contain proteins, microRNAs (miRNAs), long non-coding RNAs (lncRNA), and messenger RNA (mRNAs).[71] Jiang *et al*[34] demonstrated that the overexpression of the lncRNA ALAHM in EVs derived from LUAD cells markedly facilitates LM in a murine lung cancer model. LUAD-derived ALAHM in EVs also promotes hepatocellular secretion of hepatocyte growth factor by binding to AUF1. This event enhances LUAD cell proliferation, invasion, and migration. Consequently, ALAHM-containing EVs derived from LUAD cells increase the hepatocyte growth factor levels and promote LM of LUAD cells.

Additionally, a distinct correlation was found between LM and serum exosomal miRNAs in NSCLC.[72] miRNAs (e.g., miR-198-5p) impact the formation of mouse LMs by directly targeting fucosyltransferase 8 (FUT8) in NSCLC *in vivo*.[73] Lung cancer cell-derived exosomes containing miR-122-5p induce the EMT-related mechanisms in hepatocytes. These mechanisms include N-cadherin and vimentin expression enhancement, reduction of the expression of epithelial marker E-cadherin, and induction of liver cell migration, thereby establishing a pre-metastatic microenvironment for metastatic lung cancer.[74] Moreover, bone marrow-derived cell-secreted EVs containing miR-92a promote LM of lung cancer by potentiating HepSC activation, subsequently increasing extracellular matrix deposition in mice. This environment promotes the formation of an immunosuppressive PMN for LM by boosting the recruitment of MDSCs.[75] Mechanistically, miR-92a in EVs augments the activation of HepSCs induced via TGF-β signaling, by directly targeting the suppressor of mothers against decapentaplegic family member.

Overall, the liver microenvironment consists of various cell types, including neutrophils, macrophages, MDSCs, T cells, DCs, NK cells, and other cells. The above cell types interact with each other via cytokines/chemokines, including CCL7/CCR3, CCL5/CCR5, and TGF-β, and exosomal substances, i.e., lncRNAs and miRNAs [Figure 2]. Together, these impact the proliferation, invasion, migration, and EMT of NSCLC cells, thus influencing the establishment and growth of metastases. They represent potential targets for preventing and treating LM. However, LM is a complex process involving multiple factors. Despite the available data, liver metastatic TME is incompletely understood due to its complexity and heterogeneity. Studies to date have only preliminarily explored the underlying mechanism. An in-depth and extensive understanding of the specific mechanism using novel technologies, such as single-cell sequencing, spatial profiling technologies, and multiplex immunofluorescence, is required in future preclinical and clinical studies with large sample sizes.

Prospective Preclinical and Clinical Strategies for Targeting the Liver Metastatic Microenvironment

TME is a key factor influencing tumor treatment outcomes. LMs can promote systemic tolerance of antigen-specific T cells via immunosuppressive TME in preclinical animal models and cancer patients.[4] Different immunosuppressive phenotypes in the liver microenvironment are potential therapeutic targets in patients with LMs.[12] Although clinical data have demonstrated the therapeutic effect of immunotherapy coupled with chemotherapy, radiotherapy, and vascular endothelial growth factor (VEGF) blockade for targeting the liver TME [Table 1], limited preclinical findings illustrate the implication of immunotherapy coupled with radiotherapy and neutrophil extracellular traps (NETs) inhibitors in lung cancer patients with LMs [Table 2].

Immunotherapy

Metastatic tumors exhibit a more profound immunosuppressive microenvironment than primary tumors owing to the clonal evolution of cancer cells, immunophenotypic changes in infiltrating cells, and organ-specific ecological niches.[76] LMs systematically reduce the efficacy of immunotherapy in patients and preclinical models, while immunosuppressive macrophages are recruited to promote apoptosis of antigen-specific T cells in the liver, eventually leading to systemic loss of T cells and reduced immunotherapeutic efficacy.[4] Therefore, targeting the mechanism by which liver metastatic TME factors drive immunosuppression is an attractive strategy for improving immunotherapy of LMs. Monoclonal antibodies

Table 1: Therapeutic agents for liver metastases of NSCLC in clinical trials.

Therapeutic strategies	Therapeutic agents	Condition/population	Phase	Results	Population (LMs/all)	Study name [NCT No.]	Refs
Immunotherapy	Durvalumab	Advanced/metastatic NSCLC with LMs	I/II	Median OS: 5.5 months LM+/16.7 months LM–	96/304 patients (31.6%)	Study 1108 [NCT01693562]	[7]
	Durvalumab	Advanced/metastatic NSCLC with LMs	II	Median OS: 5.0 months LM+/10.2 months LM–	47/263 patients (17.9%)	ATLANTIC [NCT02087423]	[7,122]
	Nivolumab *vs*. docetaxel	Advanced NSCLC with LMs	III	3-year OS rate: 8% *vs*. 17%	–	CheckMate 017 and CheckMate 057	[78]
Immunotherapy plus chemotherapy	Pembrolizumab + pemetrexed-platinum *vs*. placebo + pemetrexed-platinum	Advanced nonsquamous NSCLC with LMs	III	Improved OS (22.0 months *vs*. 10.7 months) and PFS (9.0 months *vs*. 4.9 months), regardless of the LM status	–	Keynote 189 [NCT02578680]	[88]
	Atezolizumab + carboplatin + paclitaxel/carboplatin + nab-paclitaxel *vs*. carboplatin + nab-paclitaxel	Chemotherapy-naïve stage IV nonsquamous NSCLC	III	Did not achieve significant OS benefits: HR, 1.04; 95% CI, 0.63–1.72	–	IMpower 130 [NCT02367781]	[89]
	Atezolizumab + carboplatin or cisplatin + pemetrexed *vs*. carboplatin or cisplatin + pemetrexed	Chemotherapy-naïve stage IV nonsquamous NSCLC	III	Did not achieve significant OS benefits: HR, 0.99; 95% CI, 0.57–1.70	66 *vs*. 49	IMpower 132 [NCT02657434]	[90]
Immunotherapy plus radiotherapy	Durvalumab *vs*. placebo	Unresectable, stage III NSCLC and no disease progression after concurrent chemoradiotherapy	III	Estimated 5-year OS rate: 42.9% (95% CI 38.2–47.4%) *vs*. 33.4% (95% CI 27.3–39.6%)	–	PACIFIC	[99]
VEGF blockade and immunotherapy plus chemotherapy	ABCP *vs*. BCP	Metastatic nonsquamous NSCLC	III	PFS, 8.3 months *vs*. 6.8 months; median OS, 19.2 months *vs*. 14.7 months	–	IMpower150 [NCT02366143]	[105]

ABCP: Atezolizumab plus bevacizumab plus carboplatin plus paclitaxel; BCP: Bevacizumab plus carboplatin plus paclitaxel; CI: Confidence interval; HR: Hazard ratio; LM: Liver metastasis; NSCLC: Non-small cell lung cancer; OS: Overall survival; PFS: Progression-free survival; VEGF: Vascular endothelial-derived growth factor; Refs: References; –: Not available.

Table 2: Preclinical therapeutic strategies targeting TME in LM of NSCLC.

Therapeutic strategies	Therapeutic agents	Murine tumor model	Role	Function	Immunological effects	Refs
Radiation therapy	Liver-directed radiotherapy	Mice with subcutaneous tumors and liver tumors	Suppress	Metastasis	Prevented antigen-specific T-cell depletion and increased tumor antigen release and T-cell infiltration	[4]
	Stereotactic body radiation	–	Suppress	Metastasis	Promoted the recruitment and entry of activated $CD8^+$ T cells and released specific chemokines (CXCL16, CXCL9, and CXCL10) and pro-inflammatory cytokines (IFN-α and TNF-α)	[4]
	Stereotactic body radiation	–	Suppress	Metastasis	Released molecular pattern related to immunogenicity risk that attracted macrophages, primed phagocytic signals for macrophages and DCs, facilitated the death of immunogenic cells, upregulated the tumor surface histocompatibility complex class I protein, and improved the recognition of cytotoxic T cells to tumor-related antigens	[8]
Immunotherapy plus radiotherapy	Liver-directed radiotherapy plus anti-PD-L1	Mice with subcutaneous MC38 tumors and liver tumors	Suppress	Metastasis	Enhanced $Ki67^+$, IFN-γ^+, and granzyme B^+ $CD8^+$ T cells in the liver	[94]
DNase1 or NEi	DNase1 or NEi	C57BL/6 mice	Suppress	LM	Reduced cancer cell adhesion to the liver	[113]

CXCL: C-X-C motif ligand; DCs: Dendritic cells; DNase1: NET-degrading enzyme; IFN-α: Interferon-α; IFN-γ: Interferon-γ; LM: Liver metastasis; NEi: NET inhibitor; NETs: Neutrophil extracellular traps; NSCLC: Non-small cell lung cancer; PD-L1: Programed cell death ligand 1; Refs: References; TGF-β: Transforming growth factor-β; TME: Tumor microenvironment; TNF-α: Tumor necrosis factor-α; Tregs: Regulatory T cells.

targeting the immune checkpoint PD-1/PD-L1 are mainly used in the clinic, and therapeutic explorations targeting other immunosuppressive cells are also available in basic experiments. For instance, immunotherapeutic interventions targeting Treg, such as Treg-depleting CTLA-4 blockade or functional inhibition via an enhancer of zeste 2 inhibitor, restored sensitivity to the systemic antitumor immunity and ICI activity in the experimental LM model.[12] Moreover, adding MDSC blockade to PD-1 inhibitors sufficiently suppresses LMs in mice.[77]

In two retrospective analyses of Study 1108 and the ATLANTIC study, the OS and PFS were worse when LMs were present at baseline in patients with metastatic or advanced NSCLC who received the anti-PD-L1 antibody durvalumab.[7] PD-L1 was an independent predictor of treatment benefit regardless of the tumor status.[7] The median OS was 4–5 months in patients with LMs who received standard therapy or those who received the anti-PD-1/PD-L1 antibody durvalumab.[7] The latest follow-up results of the CheckMate 057 and 017 studies showed that nivolumab improved OS compared with docetaxel {median OS (95% confidence interval [CI]): 6.8 (4.9–10.4) months versus 5.9 (4.7–7.3) months; 3-year OS rate: 8% versus 17%} in 193 patients with LMs, revealing the long-term clinical benefits of nivolumab for patients with LMs and NSCLC treated previously.[78] However, the overall randomized pooled patients with squamous or nonsquamous NSCLC benefited from immunotherapy to a greater extent than pooled patients with LMs (median OS: 11.1 months *vs.* 6.8 months).[78] In the updated 5-year follow-up, a benefit was observed from nivolumab treatment, even in patients with LMs, compared with docetaxel treatment (hazard ratio [HR]: 0.67, 95% CI: 0.50–0.89), while the 5-year PFS rates were 8.0% and 0, respectively.[79] From these observations, patients with LMs of lung cancer may present a better prognosis if treated with nivolumab, although the benefits are less pronounced than those observed in lung cancer patients without LMs.

In another retrospective study, 166 patients with metastatic or advanced NSCLC were treated with ICI monotherapy (atezolizumab, pembrolizumab, or nivolumab) from March 2016 to August 2021. Treatment results revealed that LM is an early death risk factor in patients with metastatic or advanced NSCLC treated with ICIs.[80]

Given the poor efficacy of immune monotherapy in the treatment of LMs, actively exploring the possibility of immune combination therapies may be meaningful.

Immunotherapy in combination with chemotherapy

Chemotherapy can trigger immune activation by eliciting immunogenic cell death and subsequent release of tumor-associated neoantigens, which in turn activate APCs via Toll-like receptors,[81] and chemotherapy can enhance the immunogenicity of tumor cells by inducing the expression of MHC-1 molecules and tumor-specific cell surface antigens.[82] Likewise, multiple chemotherapeutic agents studied in NSCLC can indirectly promote effective anticancer immune activity via inducing immunogenic cell death.[83]

Although LMs are commonly considered a poor prognostic factor of immunotherapy, some studies have revealed that ICI combined with anti-VEGF therapy and chemotherapy or ICI monotherapy prolongs the PFS and OS of NSCLC patients with LMs.[84] Given the complexity of antitumor immunity and the positive influence of chemotherapy on immune responses, patients with LMs treated with a combination of ICIs and chemotherapy may receive a greater benefit than those treated with immunotherapy or chemotherapy alone.[85]

A meta-analysis demonstrated that in patients with LM, the administration of PD-1/PD-L1 suppressor in conjunction with chemotherapy reduced the risk of death by 21% and the risk of progression by 31% (HR: 0.79, 95% CI: 0.62–0.80 and HR: 0.69, 95% CI: 0.58–0.81, respectively).[85] The pooled HRs of PFS and OS among lung cancer patients with LMs and those without LMs were 1.11 (95% CI: 0.92–1.34) and 1.03 (95% CI: 0.80–1.35), respectively. These results suggest that this treatment is equally effective in lung cancer patients with and without LMs. Another meta-analysis of randomized controlled trials revealed that immunochemotherapy and chemotherapy as first-line treatments of advanced NSCLC led to significantly longer PFS rates in patients without LMs (HR with LM *vs*. without LM: 0.81 *vs*. 0.61, respectively, $P = 0.035$).[86] Moreover, a network meta-analysis of 1141 patients with NSCLC and LMs showed that atezolizumab combined with bevacizumab plus chemotherapy, and pembrolizumab combined with chemotherapy resulted in superior PFS compared with other immunotherapy, chemotherapy, or anti-angiogenic treatments.[87]

In the KEYNOTE-189 study (NCT02578680), patients with advanced nonsquamous NSCLC and LMs received combined chemoimmunotherapy (n = 66, active arm) or standard chemotherapy (n = 49, control arm).[88] First-line pembrolizumab combined with pemetrexed exhibited a significant increase in PFS (9.0 months *vs*. 4.9 months, HR: 0.48, 95% CI: 0.40–0.58) and OS (22.0 months *vs*. 10.7 months, HR: 0.56, 95% CI: 0.45–0.70) in metastatic nonsquamous NSCLC, regardless of the LM status, with manageable safety and tolerability.[88] Owing to these encouraging results, both the European Medicines Agency and the Food and Drug Administration have authorized the use of such combination therapy as standard therapy. Therefore, patients with lung cancer, with and without LMs, can benefit from combination immunotherapy, although the benefits achieved by those with LMs are less pronounced.

However, the effect of chemotherapy plus immunotherapy in treating patients with LMs and NSCLC remains controversial. In the IMpower 130[89]/132[90] trials, patients with NSCLC and LMs who were treated with ICIs plus chemotherapy did not gain evident increases in OS (HR: 1.04, 95% CI: 0.63–1.72 and HR: 0.99, 95% CI: 0.57–1.70, respectively). Thus, the effect of anti-PD-1/PD-L1 antibody plus chemotherapy in patients with NSCLC with LMs remains controversial and warrants further studies.

Immunotherapy in combination with radiotherapy

Irradiation alone and liver-directed radiotherapy combined with immunotherapy of LM may be particularly beneficial given the pronounced immunosuppressive characteristics of liver TMEs, especially $CD8^+$ T cell elimination, which are partially attributable to the presence of drug-resistant LSECs, DCs, and KCs.[18,91]

In different basic experiments, radiotherapy showed positive or negative effects on anti-tumor immunity in the liver. This also suggests the possibility that combining radiotherapy with immunotherapy can be mutually reinforcing or complementary for the treatment of liver metastases from lung cancer. A bias-free single-cell sequencing study reported that $CD11b^+F4/80^+$ macrophages originating from hepatic monocytes are critical mediators of antigen-specific apoptosis of $CD8^+$ T cells in the LM microenvironment through the FAS–FASL pathway in a mouse model bearing both subcutaneous and liver MC38 colorectal tumors.[4] Radiotherapy also increases the expression of the anti-inflammatory cytokine TGF-β and contributes to the infiltration of functional Tregs, thereby inhibiting the immune response within the tumor.[92] This provides the rationale for why targeting bone marrow cells and Tregs for immunosuppressive functions increases the effectiveness of radioimmunotherapy in LMs. For instance, in the mouse model bearing both subcutaneous and liver MC38 colorectal tumors, targeting LM-derived inhibitory $CD11b^+$ macrophages reverses $CD8^+$ T-cell depletion.[4] Moreover, radiotherapy plus anti-CTLA4 reduces liver metastasis, partly by decreasing the proportion of Tregs ($P < 0.05$).[93] Taken together, radiotherapy concurrently blunts immunosuppressive bone marrow cells and Tregs and enhances tumor antigen release and T-cell infiltration, thus restoring the antitumor efficacy of immunotherapy.[8]

The radiation-induced increase of histocompatibility complex class I proteins on the tumor surface may also enhance the identification of tumor-associated antigens by cytotoxic T cells.[94] Radiotherapy further facilitates the recruitment and entry of activated $CD8^+$ T lymphocytes into the tumor by releasing pro-inflammatory cytokines (interferon and tumor necrosis factor-α) and specific chemokines (C-X-C motif ligand [CXCL]10, CXCL9, and CXCL16).[94] Ionizing radiation also releases pattern-associated molecules that induce an immune response, attracting macrophages, providing phagocytic signals to DCs and macrophages, and enhancing the mortality of immunogenic cells.[94,95] Moreover, radiotherapy can alter the endothelium in the tumor bed by expressing E-selectin on endothelial cells as well as intercellular adhesion molecules,[95] and exert a systemic antitumor effect by altering the immune activity and response to non-radiotherapy.[96] Thus, liver-directed radiotherapy can simultaneously block the immunosuppressive myeloid component and Tregs, prevent the depletion of antigen-specific T-cells, and reprogram the immune microenvironment, leading to a systemic antitumor immune response.[4]

Stereotactic body radiation therapy (SBRT) is a new type of radiotherapy that delivers larger doses to smaller target lesions[97] that may improve the immune system's sensitivity and increase the effect of immunotherapy by overcoming immunosuppressive TMEs.[8] SBRT administered to LMs elicited a greater systemic immune response than SBRT administered to lung metastases. Moreover, SBRT combined with anti-CTLA-4 therapy for patients with lung metastases or LMs demonstrated clinical benefits.[98]

The PACIFIC study conducted in patients with stage III NSCLC investigated the efficacy of ICI consolidation treatment following concurrent radiotherapy and chemotherapy. The study revealed that durvalumab treatment markedly increased the patients' median OS (47.5 months *vs*. 29.1 months) and decreased the rate of LMs (0.023% *vs*. 0.034%) compared with placebo control treatment.[99] Following the combination of radiotherapy and anti-CTLA-4 or anti-CD25 monoclonal antibodies, the ratio of $CD8^+/CD4^+$ cells and level of $CD8^+$ T cells markedly increased, while the Treg, $PD1^+CD4^+$ T, and $PD1^+CD8^+$ T-cell levels decreased (P <0.05). Those effects inhibited tumor growth in regions of local irradiation and distal non-irradiation, in addition to increasing OS.[93]

Clinical research on solid tumors involving NSCLC showed that the combination of ipilimumab and high-dose stereotactic ablative body radiotherapy (50 Gy administered concurrently or sequentially in four fractions) was tolerable and had promising clinical benefits beyond the radiation field. Compared with lung irradiation, liver irradiation may be related to the activation of systemic immunity. Approximately 23% of all radiated patients experienced a partial response or disease stabilization lasting ≥6 months, with an early increase in peripheral $CD8^+$ T-cell levels and elevation in PD-1 and 4-1BB expression levels on $CD8^+$ T cells.[100] A phase I clinical trial involving patients with NSCLC revealed that concurrent treatment utilizing SBRT, ipilimumab, and nivolumab was less toxic than sequential treatment and that multisite SBRT was well-tolerated in those with extensive metastases.[101] Well-tolerated and noninvasive SBRT can markedly regulate the TME and sensitize tumors to PD-1/PD-L1 suppressors.

The remaining challenges, including the optimal radiation dose and fractionation, optimal schedule for the combination of PD-1/PD-L1 suppressors and SBRT, choice of an appropriate irradiation amount and target, and safety and efficacy evaluations, should be addressed before using this method in clinical practice.[97] Other clinical trials investigating the combination of ICIs and radiotherapy of NSCLC-derived LM are currently underway. These trials include the use of pembrolizumab with hepatic SBRT (NCT05430009), radiotherapy plus NBTXR3 (hafnium oxide-containing nanoparticles) with anti-PD-1/PD-L1 monoclonal antibodies (NCT05039632).

Immunotherapy in combination with anti-angiogenesis therapy

Angiogenic molecules, such as VEGF, promote an immunosuppressive microenvironment in NSCLC.[102] Mechanistically, VEGF promotes immunosuppression via triggering vascular abnormalities, augmenting PD-L1 expression, inhibiting antigen presentation and immune effector cells, or enhancing the immunosuppressive effect of Tregs, MDSCs, and macrophages.[103] Correspondingly, anti-angiogenesis therapy plus ICIs may normalize tumor vasculature, relieve hypoxia via augmented tumor perfusion, and enhance the activity of effector T cells, thus improving survival outcomes.[94]

In patients with LMs and NSCLC, the group treated with atezolizumab in combination with chemotherapy plus bevacizumab therapy showed the highest overall response and increased PFS rates compared with the group using dual ICIs, ICI-free treatments, and ICI plus chemotherapy.[104] Patients with LMs benefit from bevacizumab because of VEGF-A overexpression in liver lesions.[105,106] In a subgroup analysis conducted by Sandler *et al*,[107] patients with LMs significantly benefited from bevacizumab (HR: 0.68, 95% CI: 0.49–0.96) compared with those with metastases in other sites.

In the IMpower150 trial (NCT02366143), across the intention-to-treat population (comprising those with *ALK* or *EGFR* changes), PFS was longer in the atezolizumab plus bevacizumab plus carboplatin plus paclitaxel (ABCP) group than in the bevacizumab plus carboplatin plus paclitaxel (BCP) group (8.3 months *vs*. 6.8 months, HR for disease progression or death: 0.62, 95% CI: 0.52–0.74, P <0.001).[105] When patients with *ALK* or *EGFR* mutations were excluded, the median OS was greater in the ABCP group than in the BCP group (19.2 months *vs*. 14.7 months, HR for death: 0.78, 95% CI: 0.64–0.96, P = 0.02).[105] Therefore, atezolizumab, a humanized anti-PD-L1 monoclonal antibody, in combination with bevacizumab plus nab-paclitaxel and carboplatin is a worthwhile emerging first-line therapy for metastatic nonsquamous NSCLC.[108] Intriguingly, a subgroup analysis of the phase III IMpower150 trial announced at the 2019 American Society of Clinical Oncology Annual Meeting demonstrated no difference in the results between chemotherapy based on atezolizumab plus platinum and that based on bevacizumab plus platinum (5.4 months). This result implies that a combination of anti-angiogenic agents and immunotherapy may produce a survival advantage.

A meta-analysis revealed that the PFS benefit of adding atezolizumab to chemotherapy as first-line therapy was not remarkable in individuals with advanced NSCLC and LMs (HR: 0.81, 95% CI: 0.64–1.04); however, it was notable in those without LMs (HR: 0.61, 95% CI: 0.55–0.68).[109] Conversely, this difference was absent when incorporating treatment arms, including bevacizumab. Furthermore, a subgroup of patients with LMs and driver mutation-addicted tumors benefited most from the use of a combination of bevacizumab and chemotherapy. This result suggests that the metastatic location and genetic background of the tumor are key determinants of therapeutic responses.[110] Moreover, a *post hoc* analysis based on a phase III, double-blinded, multicentered randomized clinical test revealed that anlotinib leads to better PFS in patients with pretreated NSCLC and LMs. This observation suggests that anlotinib may be a possible third-line treatment for these patients.[111]

Other therapeutic strategies

NETs are DNA networks associated with neutrophil granular proteins, including neutrophil elastase (NE) and myeloperoxidase, which reportedly trap circulating cancer cells, promoting LM, as well as exogenous pathogens.[112] Decreased adhesion of cancer cells to the liver was noted in C57BL/6 mice with the NET-deficient LLC cell subline H59. This finding suggests that these cells reduce the adhesion of circulating tumor cells to liver sinusoids and LMs *in vivo*.[113] NET inhibitors (NEi: sivelestat) or a NET-degrading enzyme (DNase1) exhibit similar results in these mice, demonstrating the success of NET-targeting therapy.

Hakoda *et al*[114] reported that hepatectomy of a solitary LM from NSCLC was the preferred management for a 74-year-old man with a stable condition. In a retrospective study conducted on 2038 patients who underwent NSCLC resection between January 1997 and December 2015, hepatectomy was as effective as a multimodal treatment for an oligometastatic hepatic recurrence of NSCLC.[115] Consequently, surgical resection remains the primary curative therapeutic strategy for patients with resectable LMs.

In a retrospective study, mutation at residue 790 (T790M) in *EGFR* was reported in 11.4% of 316 patients with *EGFR*-positive NSCLC, while 4.4% of 365 patients with *EGFR*-positive NSCLC had LM,[116] suggesting a connection between LMs and *EGFR* mutations. In the *EGFR* mutation group that received first-line EGFR tyrosine kinase inhibitors (EGFR-TKIs), PFS (7.5 months *vs.* 11.8 months, $P = 0.0003$) and OS (20.8 months *vs.* 30.6 months, $P = 0.0190$) were shorter in those with LMs than in those without LMs.[117] This finding indicates that LM is a negative predictor of first-line EGFR-TKI treatment in patients with *EGFR*-mutant NSCLC.

A case report showed that osimertinib was efficacious in post-surgical LM from lung cancer expressing both *EGFR* L858R and T790M mutations.[118] The OS in patients without LMs in the osimertinib group was markedly superior to that in those in the erlotinib/gefitinib group, while no apparent difference existed among the afatinib and osimertinib groups.[119] In another study, EGFR-TKIs improved the prognosis of NSCLC patients with bone, brain, adrenal, and lung metastases compared with those of patients with LMs.[120]

Overall, the therapeutic strategies targeting the microenvironment of NSCLC LMs mainly include immunotherapy combined with chemotherapy, immunotherapy combined with radiotherapy, and immunotherapy combined with anti-angiogenesis therapy.

Challenges and Perspectives

Despite promising clinical implications, developing therapeutic strategies targeting the liver microenvironment remains challenging. First, both anti- and pro-inflammatory factors facilitate the colonization of cancer cells in the liver.[121] Given the dynamic nature of LM progression and the fact that various stages may overlap in time, the indications for application of drugs targetting pro- or anti-inflammatory factors are likely to be limited and difficult to determine.[121] Additionally, future work should focus on strategies to improve the early detection of LMs. Early detection may help to identify more patients eligible for resection, while detailed knowledge of the molecular mechanisms of LM may not only support early detection and prevention, but also promote precision therapy. Therefore, future studies should address the underlying molecular mechanisms and clinical features of LM to guide early detection and clinical precision therapy. In addition, effective testing of the many available agents during clinical trials requires broader biomarker integration and adaptive trials with accumulated outcome data to rapidly discard highly toxic and less active agents and integrate new therapeutics. Another issue of concern is the optimal timing of immunotherapy and radiotherapy for metastatic disease. Thus, further evidence is needed to determine the optimal treatment sequence, immune agent, and the best combination with radiotherapy, for example, improving local radiotherapy control and finding the optimal combination of immunotherapy and radiotherapy.

Conclusion

The liver is one of the primary metastatic sites of NSCLC; NSCLC patients with LMs have a poor prognosis. The LM is a complex multistep biological process. PMN plays a critical role in the hepatic microenvironment in promoting cancer LM. Disseminated tumor cells depend on interactions with the hepatic immune microenvironment for arrest, immune evasion, colonization, migration, and proliferation. LM-associated cells, such hepatocytes, LSECs, neutrophils, macrophages, T cells, NK cells, DCs, MDSCs, and HepSCs, and their secreted cytokines, chemokines, and growth factors, as well as the key molecules and signaling pathways represent reasonable targets for the prevention and treatment of LMs. The current preclinical models suggest immunosuppressive phenotype in LMs, such as Tregs, macrophages, and MDSCs, as the underlying culprits in LM of NSCLC and highlight that combined immunotherapy may help to counteract these immunosuppressive alterations.

Despite the availability of various treatment modalities, LM presents a worrisome prognosis and indicates the urgent need for clinical and scientific studies to help develop interventions for its prevention and treatment. Immunotherapy combined with chemotherapy, anti-angiogenesis therapy or radiotherapy might be feasible and effective interventions for patients with LMs. In summary, the incorporation of immunotherapy combinations into the LM treatment field will be the backbone of future drug development. Future studies should explore the optimal treatment timing and immunotherapy agents and optimize immunotherapy strategies to improve the outcomes of this population.

Funding

This study was supported by the National Natural Science Foundation of China (Nos. 82202989 and

82003089), the Regional Innovation Cooperation Project of the Sichuan Science and Technology Program (No. 2021YFQ0029), the China Postdoctoral Science Foundation (No. 2022M722279); the Sichuan Science and Technology Program (No. 2023YFS0163), the Postdoctoral Research Project of West China Hospital, Sichuan University, Chengdu, China (No. 2021HXBH045); Fundamental Research Funds for the Central Universities (No. 2022SCU12063); the Sichuan University Postdoctoral Interdisciplinary Innovation Fund (awarded to Lingling Zhu).

Conflict of interest

None.

References

1. Herbst RS, Morgensztern D, Boshoff C. The biology and management of non-small cell lung cancer. Nature 2018;553:446–454. doi: 10.1038/nature25183.
2. Siegel RL, Miller KD, Fuchs HE, Jemal A. Cancer Statistics, 2021. CA Cancer J Clin 2021;71:7–33. doi: 10.3322/caac.21654.
3. Fang Y, Su C. Research Progress on the Microenvironment and Immunotherapy of Advanced Non-Small Cell Lung Cancer With Liver Metastases. Front Oncol 2022;12:893716. doi: 10.3389/fonc.2022.893716.
4. Yu J, Green MD, Li S, Sun Y, Journey SN, Choi JE, *et al*. Liver metastasis restrains immunotherapy efficacy via macrophage-mediated T cell elimination. Nat Med 2021;27:152–164. doi: 10.1038/s41591-020-1131-x.
5. Riihimäki M, Hemminki A, Fallah M, Thomsen H, Sundquist K, Sundquist J, et al. Metastatic sites and survival in lung cancer. Lung Cancer 2014;86:78–84. doi: 10.1016/j.lungcan.2014.07.020.
6. Yang J, Peng A, Wang B, Gusdon AM, Sun X, Jiang G, *et al*. The prognostic impact of lymph node metastasis in patients with non-small cell lung cancer and distant organ metastasis. Clin Exp Metastasis 2019;36:457–466. doi: 10.1007/s10585-019-09985-y.
7. Sridhar S, Paz-Ares L, Liu H, Shen K, Morehouse C, Rizvi N, et al. Prognostic Significance of Liver Metastasis in Durvalumab-Treated Lung Cancer Patients. Clin Lung Cancer 2019;20:e601–e608. doi: 10.1016/j.cllc.2019.06.020.
8. Corrao G, Marvaso G, Ferrara R, Lo Russo G, Gugliandolo SG, Piperno G, et al. Stereotatic radiotherapy in metastatic non-small cell lung cancer: Combining immunotherapy and radiotherapy with a focus on liver metastases. Lung Cancer 2020;142:70–79. doi: 10.1016/j.lungcan.2020.02.017.
9. Choi MG, Choi CM, Lee DH, Kim SW, Yoon S, Kim WS, *et al*. Different prognostic implications of hepatic metastasis according to front-line treatment in non-small cell lung cancer: a real-world retrospective study. Transl Lung Cancer Res 2021;10:2551–2561. doi: 10.21037/tlcr-21-206.
10. Pawlik TM, Gleisner AL, Bauer TW, Adams RB, Reddy SK, Clary BM, *et al*. Liver-directed surgery for metastatic squamous cell carcinoma to the liver: results of a multi-center analysis. Ann Surg Oncol 2007;14:2807–2816. doi: 10.1245/s10434-007-9467-8.
11. Lee JC, Green MD, Huppert LA, Chow C, Pierce RH, Daud AI. The Liver-Immunity Nexus and Cancer Immunotherapy. Clin Cancer Res 2022;28:5–12. doi: 10.1158/1078-0432.CCR-21-1193.
12. Lee JC, Mehdizadeh S, Smith J, Young A, Mufazalov IA, Mowery CT, *et al*. Regulatory T cell control of systemic immunity and immunotherapy response in liver metastasis. Sci Immunol 2020;5:eaba0759. doi: 10.1126/sciimmunol.aba0759.
13. Funazo T, Nomizo T, Kim YH. Liver Metastasis Is Associated with Poor Progression-Free Survival in Patients with Non-Small Cell Lung Cancer Treated with Nivolumab. J Thorac Oncol 2017;12:e140–e141. doi: 10.1016/j.jtho.2017.04.027.
14. Brodt P. Role of the Microenvironment in Liver Metastasis: From Pre- to Prometastatic Niches. Clin Cancer Res 2016;22:5971–5982. doi: 10.1158/1078-0432.CCR-16-0460.
15. Thomson AW, Knolle PA. Antigen-presenting cell function in the tolerogenic liver environment. Nat Rev Immunol 2010;10:753–766. doi: 10.1038/nri2858.
16. Horst AK, Neumann K, Diehl L, Tiegs G. Modulation of liver tolerance by conventional and nonconventional antigen-presenting cells and regulatory immune cells. Cell Mol Immunol 2016;13:277–292. doi: 10.1038/cmi.2015.112.
17. Milette S, Sicklick JK, Lowy AM, Brodt P. Molecular Pathways: Targeting the Microenvironment of Liver Metastases. Clin Cancer Res 2017;23:6390–6399. doi: 10.1158/1078-0432.CCR-15-1636.
18. Tsilimigras DI, Brodt P, Clavien PA, Muschel RJ, D'Angelica MI, Endo I, et al. Liver metastases. Nat Rev Dis Primers 2021;7:27. doi: 10.1038/s41572-021-00261-6.
19. Horn SR, Stoltzfus KC, Lehrer EJ, Dawson LA, Tchelebi L, Gusani NJ, *et al*. Epidemiology of liver metastases. Cancer Epidemiol 2020;67:101760. doi: 10.1016/j.canep.2020.101760.
20. da Silva LM, da Silva GT, Bergmann A, Costa GJ, Zamboni MM, Santos Thuler LC. Impact of different patterns of metastasis in non-small-cell lung cancer patients. Future Oncol 2021;17:775–782. doi: 10.2217/fon-2020-0587.
21. Juarez-Garcia A, Sharma R, Hunger M, Kayaniyil S, Penrod JR, Chouaïd C. Real-world effectiveness of immunotherapies in pretreated, advanced non-small cell lung cancer Patients: A systematic literature review. Lung Cancer 2022;166:205–220. doi: 10.1016/j.lungcan.2022.03.008.
22. Yoshimura A, Yamada T, Tsuji T, Hamashima R, Shiotsu S, Yuba T, *et al*. Prognostic impact of pleural effusion in EGFR-mutant non-small cell lung cancer patients without brain metastasis. Thorac Cancer 2019;10:557–563. doi: 10.1111/1759-7714.12979.
23. Kitadai R, Okuma Y, Hakozaki T, Hosomi Y. The efficacy of immune checkpoint inhibitors in advanced non-small-cell lung cancer with liver metastases. J Cancer Res Clin Oncol 2020;146:777–785. doi: 10.1007/s00432-019-03104-w.
24. Raphael MJ, Karanicolas PJ. Regional Therapy for Colorectal Cancer Liver Metastases: Which Modality and When. J Clin Oncol 2022;40:2806–2817. doi: 10.1200/JCO.21.02505.
25. Wong YC, Tay SS, McCaughan GW, Bowen DG, Bertolino P. Immune outcomes in the liver: Is CD8 T cell fate determined by the environment. J Hepatol 2015;63:1005–1014. doi: 10.1016/j.jhep.2015.05.033.
26. Zhang K, Yang L, Wang J, Sun T, Guo Y, Nelson R, et al. Ubiquitin-specific protease 22 is critical to in vivo angiogenesis, growth and metastasis of non-small cell lung cancer. Cell Commun Signal 2019;17:167. doi: 10.1186/s12964-019-0480-x.
27. Jin X, Liu X, Zhang Z, Xu L. NT5DC2 suppression restrains progression towards metastasis of non-small-cell lung cancer through regulation p53 signaling. Biochem Biophys Res Commun 2020;533:354–361. doi: 10.1016/j.bbrc.2020.06.139.
28. Castañón E, Soltermann A, López I, Román M, Ecay M, Collantes M, *et al*. The inhibitor of differentiation-1 (Id1) enables lung cancer liver colonization through activation of an EMT program in tumor cells and establishment of the pre-metastatic niche. Cancer Lett 2017;402:43–51. doi: 10.1016/j.canlet.2017.05.012.
29. Román M, López I, Guruceaga E, Baraibar I, Ecay M, Collantes M, *et al*. Inhibitor of Differentiation-1 Sustains Mutant KRAS-Driven Progression, Maintenance, and Metastasis of Lung Adenocarcinoma via Regulation of a FOSL1 Network. Cancer Res 2019;79:625–638. doi: 10.1158/0008-5472.CAN-18-1479.
30. Jan YH, Lai TC, Yang CJ, Lin YF, Huang MS, Hsiao M. Adenylate kinase 4 modulates oxidative stress and stabilizes HIF-1α to drive lung adenocarcinoma metastasis. J Hematol Oncol 2019;12:12. doi: 10.1186/s13045-019-0698-5.
31. Chen B, Song Y, Zhan Y, Zhou S, Ke J, Ao W, *et al*. Fangchinoline inhibits non-small cell lung cancer metastasis by reversing epithelial-mesenchymal transition and suppressing the cytosolic ROS-related Akt-mTOR signaling pathway. Cancer Lett 2022;543:215783. doi: 10.1016/j.canlet.2022.215783.
32. Maeda S, Hikiba Y, Sakamoto K, Nakagawa H, Hirata Y, Hayakawa Y, *et al*. Ikappa B kinasebeta/nuclear factor-kappaB activation controls the development of liver metastasis by way of interleukin-6 expression. Hepatology 2009;50:1851–1860. doi: 10.1002/hep.23199.
33. Peng S, Dong W, Chu Q, Meng J, Yang H, DU Y, *et al*. Traditional Chinese Medicine Brucea Javanica Oil Enhances the Efficacy of Anlotinib in a Mouse Model of Liver-Metastasis of Small-cell Lung Cancer. In Vivo 2021;35:1437–1441. doi: 10.21873/invivo.12395.
34. Jiang C, Li X, Sun B, Zhang N, Li J, Yue S, *et al*. Extracellular vesicles promotes liver metastasis of lung cancer by ALAHM increasing hepatocellular secretion of HGF. iScience 2022;25:103984. doi: 10.1016/j.isci.2022.103984.

35. Yates C, Shepard CR, Papworth G, Dash A, Beer Stolz D, Tannenbaum S, *et al*. Novel three-dimensional organotypic liver bioreactor to directly visualize early events in metastatic progression. Adv Cancer Res 2007;97:225–246. doi: 10.1016/S0065-230X(06)97010-9.
36. Kim J, Lee C, Kim I, Ro J, Kim J, Min Y, *et al*. Three-Dimensional Human Liver-Chip Emulating Premetastatic Niche Formation by Breast Cancer-Derived Extracellular Vesicles. ACS Nano 2020;14:14971–14988. doi: 10.1021/acsnano.0c04778.
37. Clark AM, Ma B, Taylor DL, Griffith L, Wells A. Liver metastases: Microenvironments and ex-vivo models. Exp Biol Med (Maywood) 2016;241:1639–1652. doi: 10.1177/1535370216658144.
38. Oh BY, Hong HK, Lee WY, Cho YB. Animal models of colorectal cancer with liver metastasis. Cancer Lett 2017;387:114–120. doi: 10.1016/j.canlet.2016.01.048.
39. Brown ZJ, Heinrich B, Greten TF. Mouse models of hepatocellular carcinoma: an overview and highlights for immunotherapy research. Nat Rev Gastroenterol Hepatol 2018;15:536–554. doi: 10.1038/s41575-018-0033-6.
40. Wang Y, Wang F, Qin Y, Lou X, Ye Z, Zhang W, *et al*. Recent progress of experimental model in pancreatic neuroendocrine tumors: drawbacks and challenges. Endocrine 2023;80:266–282. doi: 10.1007/s12020-023-03299-6.
41. Qu J, Kalyani FS, Liu L, Cheng T, Chen L. Tumor organoids: synergistic applications, current challenges, and future prospects in cancer therapy. Cancer Commun (Lond) 2021;41:1331–1353. doi: 10.1002/cac2.12224.
42. Krüger A. Premetastatic niche formation in the liver: emerging mechanisms and mouse models. J Mol Med (Berl) 2015;93:1193–1201. doi: 10.1007/s00109-015-1342-7.
43. Yin W, Lv J, Yao Y, Zhao Y, He Z, Wang Q, et al. Elevations of monocyte and neutrophils, and higher levels of granulocyte colony-stimulating factor in peripheral blood in lung cancer patients. Thorac Cancer 2021;12:2680–2690. doi: 10.1111/1759-7714.14103.
44. McDonald B, Spicer J, Giannais B, Fallavollita L, Brodt P, Ferri LE. Systemic inflammation increases cancer cell adhesion to hepatic sinusoids by neutrophil mediated mechanisms. Int J Cancer 2009;125:1298–1305. doi: 10.1002/ijc.24409.
45. Spicer JD, McDonald B, Cools-Lartigue JJ, Chow SC, Giannias B, Kubes P, *et al*. Neutrophils promote liver metastasis via Mac-1-mediated interactions with circulating tumor cells. Cancer Res 2012;72:3919–3927. doi: 10.1158/0008-5472.CAN-11-2393.
46. Massara M, Bonavita O, Savino B, Caronni N, Mollica Poeta V, Sironi M, *et al*. ACKR2 in hematopoietic precursors as a checkpoint of neutrophil release and anti-metastatic activity. Nat Commun 2018;9:676. doi: 10.1038/s41467-018-03080-8.
47. Li P, Lu M, Shi J, Hua L, Gong Z, Li Q, et al. Dual roles of neutrophils in metastatic colonization are governed by the host NK cell status. Nat Commun 2020;11:4387. doi: 10.1038/s41467-020-18125-0.
48. Chang MM, Wu SZ, Yang SH, Wu CC, Wang CY, Huang BM. FGF9/FGFR1 promotes cell proliferation, epithelial-mesenchymal transition, M2 macrophage infiltration and liver metastasis of lung cancer. Transl Oncol 2021;14:101208. doi: 10.1016/j.tranon.2021.101208.
49. Ioannou GN. The Role of Cholesterol in the Pathogenesis of NASH. Trends Endocrinol Metab 2016;27:84–95. doi: 10.1016/j.tem.2015.11.008.
50. Tiegs G, Lohse AW. Immune tolerance: what is unique about the liver. J Autoimmun 2010;34:1–6. doi: 10.1016/j.jaut.2009.08.008.
51. Kimura Y, Inoue A, Hangai S, Saijo S, Negishi H, Nishio J, *et al*. The innate immune receptor Dectin-2 mediates the phagocytosis of cancer cells by Kupffer cells for the suppression of liver metastasis. Proc Natl Acad Sci U S A 2016;113:14097–14102. doi: 10.1073/pnas.1617903113.
52. Zhao Y, Wu T, Shao S, Shi B, Zhao Y. Phenotype, development, and biological function of myeloid-derived suppressor cells. Oncoimmunology 2016;5:e1004983. doi: 10.1080/2162402X.2015.1004983.
53. Tumeh PC, Hellmann MD, Hamid O, Tsai KK, Loo KL, Gubens MA, *et al*. Liver Metastasis and Treatment Outcome with Anti-PD-1 Monoclonal Antibody in Patients with Melanoma and NSCLC. Cancer Immunol Res 2017;5:417–424. doi: 10.1158/2326-6066.CIR-16-0325.
54. Jenne CN, Kubes P. Immune surveillance by the liver. Nat Immunol 2013;14:996–1006. doi: 10.1038/ni.2691.
55. Xia S, Guo Z, Xu X, Yi H, Wang Q, Cao X. Hepatic microenvironment programs hematopoietic progenitor differentiation into regulatory dendritic cells, maintaining liver tolerance. Blood 2008;112:3175–3185. doi: 10.1182/blood-2008-05-159921.
56. Hiratsuka S, Tomita T, Mishima T, Matsunaga Y, Omori T, Ishibashi S, *et al*. Hepato-entrained B220(+)CD11c(+)NK1.1(+) cells regulate pre-metastatic niche formation in the lung. EMBO Mol Med 2018;10:e8643. doi: 10.15252/emmm.201708643.
57. Lee JW, Stone ML, Porrett PM, Thomas SK, Komar CA, Li JH, *et al*. Hepatocytes direct the formation of a pro-metastatic niche in the liver. Nature 2019;567:249–252. doi: 10.1038/s41586-019-1004-y.
58. Wilkinson AL, Qurashi M, Shetty S. The Role of Sinusoidal Endothelial Cells in the Axis of Inflammation and Cancer Within the Liver. Front Physiol 2020;11:990. doi: 10.3389/fphys.2020.00990.
59. Auerbach R, Lu WC, Pardon E, Gumkowski F, Kaminska G, Kaminski M. Specificity of adhesion between murine tumor cells and capillary endothelium: an in vitro correlate of preferential metastasis in vivo. Cancer Res 1987;47:1492–1496.
60. Wu W, Liao H, Ye W, Li X, Zhang J, Bu J. Fatty liver is a risk factor for liver metastasis in Chinese patients with non-small cell lung cancer. PeerJ 2019;7:e6612. doi: 10.7717/peerj.6612.
61. Mikuriya Y, Tashiro H, Kuroda S, Nambu J, Kobayashi T, Amano H, *et al*. Fatty liver creates a pro-metastatic microenvironment for hepatocellular carcinoma through activation of hepatic stellate cells. Int J Cancer 2015;136:E3–13. doi: 10.1002/ijc.29096.
62. Gowing SD, Chow SC, Cools-Lartigue JJ, Chen CB, Najmeh S, Goodwin-Wilson M, *et al*. Gram-Negative Pneumonia Augments Non-Small Cell Lung Cancer Metastasis through Host Toll-like Receptor 4 Activation. J Thorac Oncol 2019;14:2097–2108. doi: 10.1016/j.jtho.2019.07.023.
63. Ying X, Ma N, Zhang X, Guo H, Liu Y, Chen B, *et al*. Research progress on the molecular mechanisms of hepatic metastasis in lung cancer: a narrative review. Ann Palliat Med 2021;10:4806–4822. doi: 10.21037/apm-20-1675.
64. Chen L, Zheng H, Yu X, Liu L, Li H, Zhu H, *et al*. Tumor-Secreted GRP78 Promotes the Establishment of a Pre-metastatic Niche in the Liver Microenvironment. Front Immunol 2020;11:584458. doi: 10.3389/fimmu.2020.584458.
65. Stefanescu C, Van Gogh M, Roblek M, Heikenwalder M, Borsig L. TGFβ Signaling in Myeloid Cells Promotes Lung and Liver Metastasis Through Different Mechanisms. Front Oncol 2021;11:765151. doi: 10.3389/fonc.2021.765151.
66. Khan GJ, Sun L, Abbas M, Naveed M, Jamshaid T, Baig M, *et al*. In-vitro Pre-Treatment of Cancer Cells with TGF-β1: A Novel Approach of Tail Vein Lung Cancer Metastasis Mouse Model for Anti-Metastatic Studies. Curr Mol Pharmacol 2019;12:249–260. doi: 10.2174/1874467212666190306165703.
67. Tauriello D, Palomo-Ponce S, Stork D, Berenguer-Llergo A, Badia-Ramentol J, Iglesias M, *et al*. TGFβ drives immune evasion in genetically reconstituted colon cancer metastasis. Nature 2018;554:538–543. doi: 10.1038/nature25492.
68. Marvin DL, Heijboer R, Ten Dijke P, Ritsma L. TGF-β signaling in liver metastasis. Clin Transl Med 2020;10:e160. doi: 10.1002/ctm2.160.
69. Zhou H, Liu Z, Wang Y, Wen X, Amador EH, Yuan L, *et al*. Colorectal liver metastasis: molecular mechanism and interventional therapy. Signal Transduct Target Ther 2022;7:70. doi: 10.1038/s41392-022-00922-2.
70. Vaniotis G, Rayes RF, Qi S, Milette S, Wang N, Perrino S, *et al*. Collagen IV-conveyed signals can regulate chemokine production and promote liver metastasis. Oncogene 2018;37:3790–3805. doi: 10.1038/s41388-018-0242-z.
71. Li S, Qu Y, Liu L, Wang C, Yuan L, Bai H, *et al*. Tumour-derived exosomes in liver metastasis: A Pandora's box. Cell Prolif 2023;56:e13452. doi: 10.1111/cpr.13452.
72. Wang J, Xue H, Zhu Z, Gao J, Zhao M, Ma Z. Expression of serum exosomal miR-23b-3p in non-small cell lung cancer and its diagnostic efficacy. Oncol Lett 2020;20:30. doi: 10.3892/ol.2020.11891.
73. Wang S, Zhang X, Yang C, Xu S. MicroRNA-198-5p inhibits the migration and invasion of non-small lung cancer cells by targeting fucosyltransferase 8. Clin Exp Pharmacol Physiol 2019;46:955–967. doi: 10.1111/1440-1681.13154.
74. Li C, Qin F, Wang W, Ni Y, Gao M, Guo M, *et al*. hnRNPA2B1-Mediated Extracellular Vesicles Sorting of miR-122-5p Potentially

Promotes Lung Cancer Progression. Int J Mol Sci 2021;22:12866. doi: 10.3390/ijms222312866.

75. Hsu YL, Huang MS, Hung JY, Chang WA, Tsai YM, Pan YC, *et al*. Bone-marrow-derived cell-released extracellular vesicle miR-92a regulates hepatic pre-metastatic niche in lung cancer. Oncogene 2020;39:739–753. doi: 10.1038/s41388-019-1024-y.
76. Zou Y, Hu X, Zheng S, Yang A, Li X, Tang H, *et al*. Discordance of immunotherapy response predictive biomarkers between primary lesions and paired metastases in tumours: A systematic review and meta-analysis. EBioMedicine 2021;63:103137. doi: 10.1016/j.ebiom.2020.103137.
77. Conche C, Finkelmeier F, Pešić M, Nicolas AM, Böttger TW, Kennel KB, *et al*. Combining ferroptosis induction with MDSC blockade renders primary tumours and metastases in liver sensitive to immune checkpoint blockade. Gut 2023;72:1774–1782. doi: 10.1136/gutjnl-2022-327909.
78. Vokes EE, Ready N, Felip E, Horn L, Burgio MA, Antonia SJ, *et al*. Nivolumab versus docetaxel in previously treated advanced non-small-cell lung cancer (CheckMate 017 and CheckMate 057): 3-year update and outcomes in patients with liver metastases. Ann Oncol 2018;29:959–965. doi: 10.1093/annonc/mdy041.
79. Borghaei H, Gettinger S, Vokes EE, Chow L, Burgio MA, de Castro Carpeno J, *et al*. Five-Year Outcomes From the Randomized, Phase III Trials CheckMate 017 and 057: Nivolumab Versus Docetaxel in Previously Treated Non-Small-Cell Lung Cancer. J Clin Oncol 2021;39:723–733. doi: 10.1200/JCO.20.01605.
80. Takeuchi E, Kondo K, Okano Y, Kunishige M, Kondo Y, Kadota N, *et al*. Early mortality factors in immune checkpoint inhibitor monotherapy for advanced or metastatic non-small cell lung cancer. J Cancer Res Clin Oncol 2023;149:3139–3147. doi: 10.1007/s00432-022-04215-7.
81. Liang Z, Cui X, Yang L, Hu Q, Li D, Zhang X, *et al*. Co-assembled nanocomplexes of peptide neoantigen Adpgk and Toll-like receptor 9 agonist CpG ODN for efficient colorectal cancer immunotherapy. Int J Pharm 2021;608:121091. doi: 10.1016/j.ijpharm.2021.121091.
82. Galluzzi L, Humeau J, Buqué A, Zitvogel L, Kroemer G. Immunostimulation with chemotherapy in the era of immune checkpoint inhibitors. Nat Rev Clin Oncol 2020;17:725–741. doi: 10.1038/s41571-020-0413-z.
83. Mathew M, Enzler T, Shu CA, Rizvi NA. Combining chemotherapy with PD-1 blockade in NSCLC. Pharmacol Ther 2018;186:130–137. doi: 10.1016/j.pharmthera.2018.01.003.
84. Yang K, Li J, Bai C, Sun Z, Zhao L. Efficacy of Immune Checkpoint Inhibitors in Non-small-cell Lung Cancer Patients With Different Metastatic Sites: A Systematic Review and Meta-Analysis. Front Oncol 2020;10:1098. doi: 10.3389/fonc.2020.01098.
85. Qin BD, Jiao XD, Liu J, Liu K, He X, Wu Y, *et al*. The effect of liver metastasis on efficacy of immunotherapy plus chemotherapy in advanced lung cancer. Crit Rev Oncol Hematol 2020;147:102893. doi: 10.1016/j.critrevonc.2020.102893.
86. Li ZQ, Yan HC, Gu JJ, Yang YL, Zhang MK, Fang XJ. Comparative efficacy and safety of PD-1/PD-L1 Inhibitors versus platinum-based chemotherapy for the first-line treatment of advanced non-small cell lung cancer: a meta analysis of randomized controlled trials. Pharmacol Res 2020;160:105194. doi: 10.1016/j.phrs.2020.105194.
87. Yin Q, Dai L, Sun R, Ke P, Liu L, Jiang B. Clinical Efficacy of Immune Checkpoint Inhibitors in Non-Small Cell Lung Cancer Patients with Liver Metastases: A Network Meta-Analysis of Nine Randomized Controlled Trials. Cancer Res Treat 2022;54:803–816. doi: 10.4143/crt.2021.764.
88. Gadgeel S, Rodríguez-Abreu D, Speranza G, Esteban E, Felip E, Dómine M, *et al*. Updated Analysis From KEYNOTE-189: Pembrolizumab or Placebo Plus Pemetrexed and Platinum for Previously Untreated Metastatic Nonsquamous Non-Small-Cell Lung Cancer. J Clin Oncol 2020;38:1505–1517. doi: 10.1200/JCO.19.03136.
89. West H, McCleod M, Hussein M, Morabito A, Rittmeyer A, Conter HJ, *et al*. Atezolizumab in combination with carboplatin plus nab-paclitaxel chemotherapy compared with chemotherapy alone as first-line treatment for metastatic non-squamous non-small-cell lung cancer (IMpower130): a multicentre, randomised, open-label, phase 3 trial. Lancet Oncol 2019;20:924–937. doi: 10.1016/S1470-2045(19)30167-6.
90. Nishio M, Barlesi F, West H, Ball S, Bordoni R, Cobo M, *et al*. Atezolizumab Plus Chemotherapy for First-Line Treatment of Nonsquamous NSCLC: Results From the Randomized Phase 3 IMpower132 Trial. J Thorac Oncol 2021;16:653–664. doi: 10.1016/j.jtho.2020.11.025.
91. Rudqvist NP, Pilones KA, Lhuillier C, Wennerberg E, Sidhom JW, Emerson RO, *et al*. Radiotherapy and CTLA-4 Blockade Shape the TCR Repertoire of Tumor-Infiltrating T Cells. Cancer Immunol Res 2018;6:139–150. doi: 10.1158/2326-6066.CIR-17-0134.
92. Muroyama Y, Nirschl TR, Kochel CM, Lopez-Bujanda Z, Theodros D, Mao W, *et al*. Stereotactic Radiotherapy Increases Functionally Suppressive Regulatory T Cells in the Tumor Microenvironment. Cancer Immunol Res 2017;5:992–1004. doi: 10.1158/2326-6066.CIR-17-0040.
93. Ji D, Song C, Li Y, Xia J, Wu Y, Jia J, *et al*. Combination of radiotherapy and suppression of Tregs enhances abscopal antitumor effect and inhibits metastasis in rectal cancer. J Immunother Cancer 2020;8:e000826. doi: 10.1136/jitc-2020-000826.
94. Zhu L, Yu X, Wang L, Liu J, Qu Z, Zhang H, *et al*. Angiogenesis and immune checkpoint dual blockade in combination with radiotherapy for treatment of solid cancers: opportunities and challenges. Oncogenesis 2021;10:47. doi: 10.1038/s41389-021-00335-w.
95. Sharp CD, Jawahar A, Warren AC, Elrod JW, Nanda A, Alexander JS. Gamma knife irradiation increases cerebral endothelial expression of intercellular adhesion molecule 1 and E-selectin. Neurosurgery 2003;53:154–160; discussion 160–161. doi: 10.1227/01.neu.0000068840.84484.da.
96. Ozpiskin OM, Zhang L, Li JJ. Immune targets in the tumor microenvironment treated by radiotherapy. Theranostics 2019;9:1215–1231. doi: 10.7150/thno.32648.
97. Chen Y, Gao M, Huang Z, Yu J, Meng X. SBRT combined with PD-1/PD-L1 inhibitors in NSCLC treatment: a focus on the mechanisms, advances, and future challenges. J Hematol Oncol 2020;13:105. doi: 10.1186/s13045-020-00940-z.
98. Robin TP, Raben D, Schefter TE. A Contemporary Update on the Role of Stereotactic Body Radiation Therapy (SBRT) for Liver Metastases in the Evolving Landscape of Oligometastatic Disease Management. Semin Radiat Oncol 2018;28:288–294. doi: 10.1016/j.semradonc.2018.06.009.
99. Spigel DR, Faivre-Finn C, Gray JE, Vicente D, Planchard D, Paz-Ares L, et al. Five-Year Survival Outcomes From the PACIFIC Trial: Durvalumab After Chemoradiotherapy in Stage III Non-Small-Cell Lung Cancer. J Clin Oncol 2022;40:1301–1311. doi: 10.1200/JCO.21.01308.
100. Tang C, Welsh JW, de Groot P, Massarelli E, Chang JY, Hess KR, *et al*. Ipilimumab with Stereotactic Ablative Radiation Therapy: Phase I Results and Immunologic Correlates from Peripheral T Cells. Clin Cancer Res 2017;23:1388–1396. doi: 10.1158/1078-0432.CCR-16-1432.
101. Bestvina CM, Pointer KB, Karrison T, Al-Hallaq H, Hoffman PC, Jelinek MJ, *et al*. A Phase 1 Trial of Concurrent or Sequential Ipilimumab, Nivolumab, and Stereotactic Body Radiotherapy in Patients With Stage IV NSCLC Study. J Thorac Oncol 2022;17:130–140. doi: 10.1016/j.jtho.2021.08.019.
102. Zhao Y, Chen G, Chen J, Zhuang L, Du Y, Yu Q, *et al*. AK112, a novel PD-1/VEGF bispecific antibody, in combination with chemotherapy in patients with advanced non-small cell lung cancer (NSCLC): an open-label, multicenter, phase II trial. EClinicalMedicine 2023;62:102106. doi: 10.1016/j.eclinm.2023.102106.
103. Ren S, Xiong X, You H, Shen J, Zhou P. The Combination of Immune Checkpoint Blockade and Angiogenesis Inhibitors in the Treatment of Advanced Non-Small Cell Lung Cancer. Front Immunol 2021;12:689132. doi: 10.3389/fimmu.2021.689132.
104. Sheng L, Gao J, Xu Q, Zhang X, Huang M, Dai X, *et al*. Selection of optimal first-line immuno-related therapy based on specific pathological characteristics for patients with advanced driver-gene wild-type non-small cell lung cancer: a systematic review and network meta-analysis. Ther Adv Med Oncol 2021;13:17588359211018537. doi: 10.1177/17588359211018537.
105. Socinski MA, Jotte RM, Cappuzzo F, Orlandi F, Stroyakovskiy D, Nogami N, *et al*. Atezolizumab for First-Line Treatment of Metastatic Nonsquamous NSCLC. N Engl J Med 2018;378:2288–2301. doi: 10.1056/NEJMoa1716948.
106. Seo Y, Baba H, Fukuda T, Takashima M, Sugimachi K. High expression of vascular endothelial growth factor is associated with liver metastasis and a poor prognosis for patients with ductal pancreatic adenocarcinoma. *Cancer* 2000;88:2239–2245. doi: 10.1002/

(sici)1097-0142(20000515)88:10<2239::aid-cncr6>3.0.co;2-v.
107. Sandler A, Gray R, Perry MC, Brahmer J, Schiller JH, Dowlati A, *et al*. Paclitaxel-carboplatin alone or with bevacizumab for non-small-cell lung cancer. N Engl J Med 2006;355:2542–2550. doi: 10.1056/NEJMoa061884.
108. Dhillon S, Syed YY. Atezolizumab First-Line Combination Therapy: A Review in Metastatic Nonsquamous NSCLC. Target Oncol 2019;14:759–768. doi: 10.1007/s11523-019-00686-w.
109. Dafni U, Tsourti Z, Vervita K, Peters S. Immune checkpoint inhibitors, alone or in combination with chemotherapy, as first-line treatment for advanced non-small cell lung cancer. A systematic review and network meta-analysis. Lung Cancer 2019;134:127–140. doi: 10.1016/j.lungcan.2019.05.029.
110. Cantelmo AR, Dejos C, Kocher F, Hilbe W, Wolf D, Pircher A. Angiogenesis inhibition in non-small cell lung cancer: a critical appraisal, basic concepts and updates from American Society for Clinical Oncology 2019. Curr Opin Oncol 2020;32:44–53. doi: 10.1097/CCO.0000000000000591.
111. Shen Y, Lu J, Hu F, Qian J, Zhang X, Zhong R, *et al*. Effect and outcomes analysis of anlotinib in non-small cell lung cancer patients with liver metastasis: results from the ALTER 0303 phase 3 randomized clinical trial. J Cancer Res Clin Oncol 2023;149:1417–1424. doi: 10.1007/s00432-022-03964-9.
112. Cools-Lartigue J, Spicer J, McDonald B, Gowing S, Chow S, Giannias B, *et al*. Neutrophil extracellular traps sequester circulating tumor cells and promote metastasis. J Clin Invest 2013;123:3446–3458. doi: 10.1172/JCI67484.
113. Rayes RF, Mouhanna JG, Nicolau I, Bourdeau F, Giannias B, Rousseau S, *et al*. Primary tumors induce neutrophil extracellular traps with targetable metastasis promoting effects. *JCI Insight* 2019;5:e128008. doi: 10.1172/jci.insight.128008.
114. Hakoda H, Sekine Y, Ichimura H, Ueda K, Aoki S, Mishima H, *et al*. Hepatectomy for rapidly growing solitary liver metastasis from non-small cell lung cancer: a case report. *Surg Case Rep* 2019;5:71. doi: 10.1186/s40792-019-0633-6.
115. Ishige F, Takayama W, Chiba S, Hoshino I, Arimitsu H, Yanagibashi H, *et al*. Hepatectomy for oligo-recurrence of non-small cell lung cancer in the liver. Int J Clin Oncol 2018;23:647–651. doi: 10.1007/s10147-018-1262-y.
116. Chen Y, Deng J, Liu Y, Wang H, Zhao S, He Y, *et al*. Analysis of metastases in non-small cell lung cancer patients with epidermal growth factor receptor mutation. Ann Transl Med 2021;9:206. doi: 10.21037/atm-20-2925.
117. Jiang T, Cheng R, Zhang G, Su C, Zhao C, Li X, *et al*. Characterization of Liver Metastasis and Its Effect on Targeted Therapy in EGFR-mutant NSCLC: A Multicenter Study. Clin Lung Cancer 2017;18:631–639.e2. doi: 10.1016/j.cllc.2017.04.015.
118. Sakaizawa T, Toishi M, Ozawa K, Nishimura H, Maruno T, Suzuki Y. Osimertinib Was Effective for Liver Metastasis after Surgery for Lung Cancer with Simultaneous Expression of EGFR L858Ra nd T790M Mutations–A Case Report (in Japanese). Gan To Kagaku Ryoho 2019;46:1291–1293.
119. Gen S, Tanaka I, Morise M, Koyama J, Kodama Y, Matsui A, *et al*. Clinical efficacy of osimertinib in EGFR-mutant non-small cell lung cancer with distant metastasis. BMC Cancer 2022;22:654. doi: 10.1186/s12885-022-09741-8.
120. Meng C, Ge X, Tian J, Wei J, Zhao L. Prognostic role of targeted therapy in patients with multiple-site metastases from non-small-cell lung cancer. Future Oncol 2020;16:1957–1967. doi: 10.2217/fon-2020-0289.
121. Li X, Ramadori P, Pfister D, Seehawer M, Zender L, Heikenwalder M. The immunological and metabolic landscape in primary and metastatic liver cancer. Nat Rev Cancer 2021;21:541–557. doi: 10.1038/s41568-021-00383-9.
122. Garassino MC, Cho BC, Kim JH, Mazières J, Vansteenkiste J, Lena H, *et al*. Durvalumab as third-line or later treatment for advanced non-small-cell lung cancer (ATLANTIC): an open-label, single-arm, phase 2 study. Lancet Oncol 2018;19:521–536. doi: 10.1016/S1470-2045(18)30144-X.

How to cite this article: Zhu LL, Yu XZ, Tang XJ, Hu CG, Wu L, Liu YY, Zhou QH. Evolving landscape of treatments targeting the microenvironment of liver metastases in non-small cell lung cancer. Chin Med J 2024;137:1019–1032. doi: 10.1097/CM9.0000000000002981